molecules.They contain unusual bases such as methylated adenine, thymine, dihydrouracil and pseudouridine. These unusual bases are unique to tRNA. Intra-chain base pairing is observed with many base pairs, while some bases are not involved in base pairing, resulting in loops and arms formation in tRNA. These foldings in the primary structure generates a clover-leaf secondary structure (Figure 7.10).

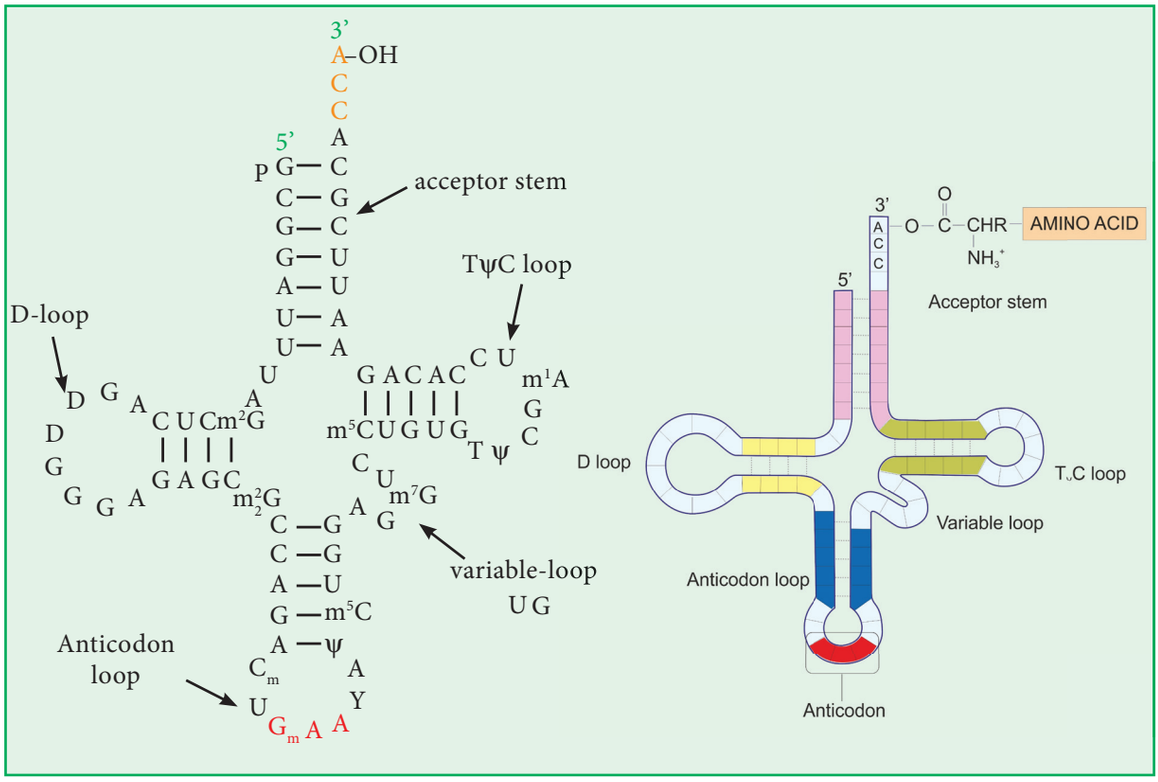


Figure 7.10 Structure of tRNA

The important features of the clover-leaf structure are,

1. An acceptor arm with base sequence "CCA" is present at the 3′end. The hydroxyl group of adenosine moiety of tRNA is responsible for getting amino acylated.
2. An anticodon arm which recognises codon on mRNA is present.
3. A TψC loop which contain an unusual base pseudouracil.
4. A D - arm which contain many dihydrouracil residues.

Functions

1. It is the carrier of amino acids to the site of protein synthesis.
2. There is at least one t-RNA molecule to each of 20 amino acids required for protein synthesis.

iii. Ribosomal RNA

Ribosomal RNA accounts for 80% of the total cellular RNA. It is present in ribosomes in combination with proteins. It is known as ribonucleoprotein. The length

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